

Eco 3271 / 3272: Engineering Economics

Midterm Lecture Notes Transcription

Lecture 1 14.07.2020

Origin: Greek word "Oikonomia" থেকে (from) Economics.

Meaning: Management of household.

Adam Smith (1776)

Published a book named: *"An enquiry in the nature and the causes of the wealth of nations."*

Definition: Economics is the science which studies the nature and the causes of the wealth of nation.

Chapter 1: Definition and Subject Matters of Economics

Lecture-2: 21.07.2020

1. Adam Smith

- **Year:** 1776
- **Book:** "An enquiry in the nature and the causes of the wealth of nations."
- **Definition:** Economics is a science of wealth.

Features of Definition:

1. Science of Wealth.
2. Emphasis on production.
3. Supports free marketing (Demand and Supply).
4. Said govt interference is not needed in economy.

Criticisms:

1. Ignores human welfare.
2. Too materialistic definition (বস্তুগত).
3. Narrow definition.

2. Alfred Marshall

- **Year:** 1890

3. Lionel Robbins

- **Year:** 1932
- **Book:** "An Essay on the nature and significance of Economic Science"
- **Definition:** Economics is a science which studies human behavior as a relationship between ends (unlimited human wants - অসীম) and scarce means (limited resource - দুস্পাপ্য) which have alternative uses.

Features:

1. Scarcity is the central problem.
2. Resources have alternative uses.
3. Choice is necessary.
4. Efficient allocation of resource.

Criticisms:

1. Don't mention technology.
2. Not suitable during unemployment.

Note: If asked to "Define Economics" in an exam, answer with a minimum of two definitions.

Subject Matters of Economics

1. **Production (Output):** Production is the process of using resources like labor, capital, and raw materials to make goods and services. This area helps us understand how businesses decide what to make and how much to produce. Focuses on transforming inputs into useful outputs to satisfy human needs.
2. **Exchange:** Exchange involves the buying and selling of goods and services between people, businesses or countries. This subject explains how prices are determined through the forces of supply and demand. Usually takes place in markets using money as a medium of trade.
3. **Distribution:** Distribution deals with how total income and wealth are divided among individuals and factors of production. The goal is to understand how fairly or efficiently economic rewards reach different parts of society.
4. **Consumption:** Consumption is the act of using goods and services to satisfy immediate human wants and needs. It represents the final stage of economic activity and drives overall economic demand. Studying consumption helps us to see how consumers choose to spend or save their money.
5. **Employment:** Employment focuses on how an economy uses its available labor for producing goods and services. It looks at job creation, working conditions and the causes of unemployment. High employment is crucial because of generating income for individuals.
6. **Economic Growth and Development:** Economic growth refers to an increase in a country's

Features	Microeconomics	Macroeconomics
1. Definition	A branch of economics which studies individual economic behavior (such as individual consumers, producers, market etc.)	A branch of economics which studies the economy as a whole (such as national output, national employment, Aggregate Demand and Supply, price level.)
2. Unit of Analysis	Individual consumers, producers, market.	Aggregate Demand, Aggregate Supply.
3. Scope	Narrow	Broad
4. Objectives	Efficient allocation of resources	Economic Stability
5. Price & 6. Income	Individual commodities price, individual Income	General Price level, National income (GDP, GNP)
7. Market & 8. Policy Focus	Individual market. Policy of single consumer / producers	Overall market economy, fiscal policy and monetary policy
9. Equilibriums	Partial equilibrium	General equilibrium
10. Decision Makers	Individual consumers, producers	Government, Central Bank (C.B.)

Central Economics Problem of Every Society

1. What to produce:

- Consumer goods/products
- Capital goods (machinery/money)

2. How to produce:

- Land (fixed), Capital, Labor.
- If Capital > Labor → Capital intensive production.
- If Labor > Capital → Labor intensive production.

3. Whom to produce:

- Rich
- Poor

Definition: Economic System is the set of organizational arrangements and institutions that are established to solve economic problems.

1. Capitalist Economic System (C.E.S)

An economic system in which govt. does not control economic activity and all problems are solved by price mechanism is known as Capitalist Economic System. *Countries: USA, UK, Singapore.*

Features:

1. Private Property (Individual, businessman, company).
2. Profit motive (maximization of profit).
3. Market economy (Competition). Demand (D) = Supply (S) → Equilibrium Price and Quantity.
4. Consumer Sovereignty/freedom.
5. Limited govt. role (No govt. intervention here).

2. Socialist Economic System (S.E.S)

An economic system in which government decides how economic resources will be allocated and means of production are owned by the government. *Countries: Soviet Union, Cuba, North Korea.*

Features:

1. Public Ownership (property, means of production).
2. Central Planning.
3. No profit motive (focus on social development).
4. Equal distribution of income.
5. Limited consumer choices.
6. Price is fixed by the govt.

3. Mixed Economic System (M.E.S)

An economic system in which most economic decisions result from the interacting of buyers and sellers in the market, but govt. plays a significant role in the allocation of resources is known as M.E.S.

Features:

1. Coexistence of public and private sectors.
2. The state sometimes regulates the economy.
3. The govt. sets development goals.

Chapter 2: Demand

Lecture-5: 04.08.2020

Quantity Demand: The amount of a good or services that a consumer is willing to and able to purchase at a given price.

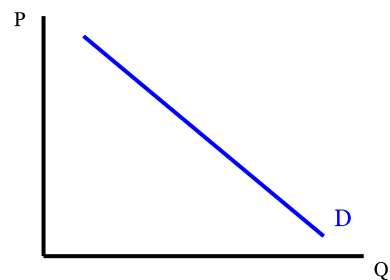
ceteris paribus - Economics term meaning "holding everything else constant".

Law of Demand: Holding everything else constant, when the price of a good or service falls, the quantity demanded of the good or service will increase and when the price of a good or service increases the quantity demanded decrease.

Demand Schedule (চাহিদা সূচী): A table that shows the relationship between the price of a product and the quantity of the product is called Demand Schedule.

Price (\$)	Quantity
175	3
150	4
125	5
100	6
75	7

Demand Curve (চাহিদা রেখা): A curve that shows the relationship between the price of a product and the quantity demanded of the product. *Note: Demand curve কখনোই X/Y অক্ষকে স্পর্শ করবে না। (Never touches X/Y axis).*



Demand Equation

$$Q_d = a - bP$$

- Q_d = Quantity Demanded (dependent variable)
- a = Constant
- $-b$ = Constant that shows slope of demand curve (downward sloping)
- P = Price (independent variable)

Variables that Shift Demand Curve: (Components determining quantity demanded, except price)

1. Income (If Income increases, quantity demanded increases).
2. Taste.
3. population
4. expected future price.
5. price of related goods.

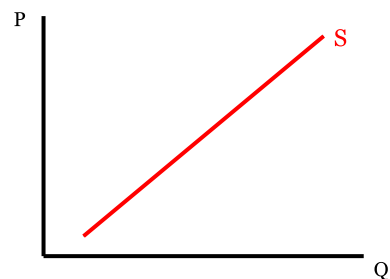
Definition: The amount of good or service that a firm or producer is willing and able to supply at a given price is called quantity supplied/supply.

Law of Supply: Holding everything else constant, increases in price cause increases in quantity supplied and vice-versa, ceteris paribus.

Supply Schedule: The table that shows the relationship between the price of product and the quantity supplied of the product is called supply schedule.

Price (\$)	Quantity
175	7
150	6
125	5
100	4
75	3

Supply Curve: A curve that shows the relationship between the price of product and the quantity supplied of the product. It has an upward sloping positive ratio.



Supply Equation

$$Q_s = c + dP \quad \text{OR} \quad Q_s = -c + dP$$

- Q_s = Quantity Supply (dependent)
- c = Constant
- d = Constant showing positive slope of supply curve
- P = Price (independent)

Variables that Shift Supply Curve:

1. Price of inputs (Land, Labor, capital).
2. Technological change.
3. Expected future price.
4. Number of firms in the market.

Note: Decrease in supply shifts curve upwards to the left. Increase in supply shifts curve

Market equilibrium is a situation in which quantity demanded equals quantity supplied.

A market equilibrium graph plots Price on the vertical Y-axis against Quantity (Q) on the horizontal X-axis. It features a downward-sloping Demand Curve (D) and an upward-sloping Supply Curve (S) which intersect at Equilibrium Point (E). This intersection determines the Equilibrium Price (P_0) and Equilibrium Quantity (Q_0), where quantity demanded equals quantity supplied ($Q_d = Q_s$) and the market is balanced.

Any price above P_0 creates a surplus (excess supply) that forces prices down, while any price below P_0 causes a shortage (excess demand) that pushes prices back up to equilibrium.

Mathematical Problem 1

Given: $Q_d = 80 - 3P$ and $Q_s = -10 + 2P$. Find equilibrium price and quantity.

Solution:

In equilibrium, $Q_d = Q_s$

$$80 - 3P = -10 + 2P$$

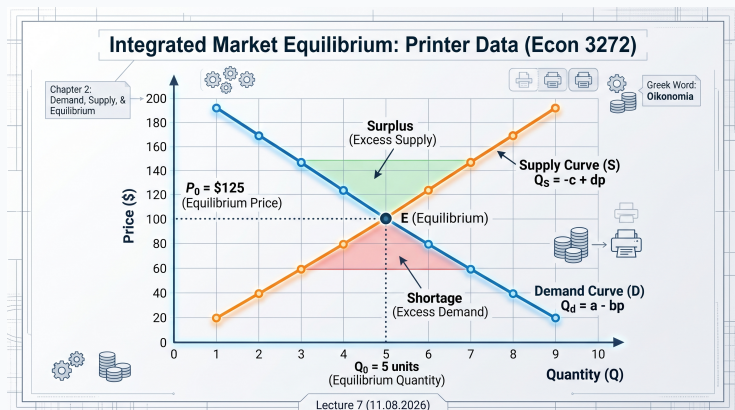
$$2P + 3P = 80 + 10$$

$$5P = 90 \Rightarrow P = 18$$

Putting $P = 18$ in the demand equation:

$$Q_d = 80 - 3(18) = 80 - 54 = 26$$

Equilibrium Price = 18, Quantity = 26.



Mathematical Problem 2

Given: $Q_d = 1200 - 10P$ and $Q_s = -400 + 6P$. Find equilibrium price and quantity.

Solution:

In equilibrium, $Q_d = Q_s$

$$1200 - 10P = -400 + 6P$$

$$16P = 1600 \Rightarrow P = 100$$

Putting $P = 100$ in the supply equation:

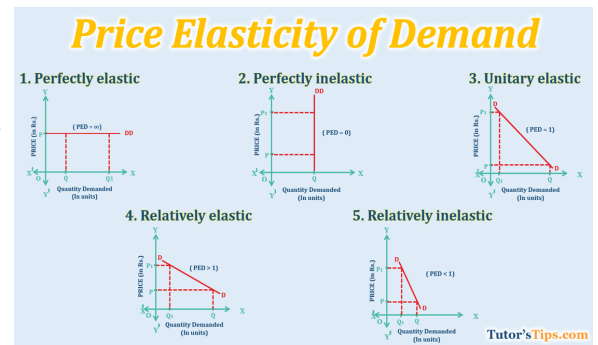
Definition: Responsiveness of dependent variable due to change in independent variable.

Types of Elasticity of Demand:

1. **Price Elasticity of Demand:** Percentage change in quantity demanded due to percentage change in price.
Math: $(\Delta Q / \Delta P) \times (P / Q)$
2. **Income Elasticity of Demand:** Percentage change in quantity demanded due to percentage change in income.
Math: $(\Delta Q / \Delta I) \times (I / Q)$
3. **Cross Price Elasticity:** Percentage change in quantity demanded of one goods due to percentage change in price of another goods.

Elasticity of Demand Curves:

- **Elastic Demand ($E_d > 1$):** Flatter downward curve.
- **Inelastic Demand ($E_d < 1$):** Steeper downward curve.
- **Unit Elasticity ($E_d = 1$):** Proportional curve.
- **Zero Elasticity ($E_d = 0$):** Perfectly vertical line.
- **Infinite Elasticity ($E_d = \infty$):** Perfectly horizontal line.

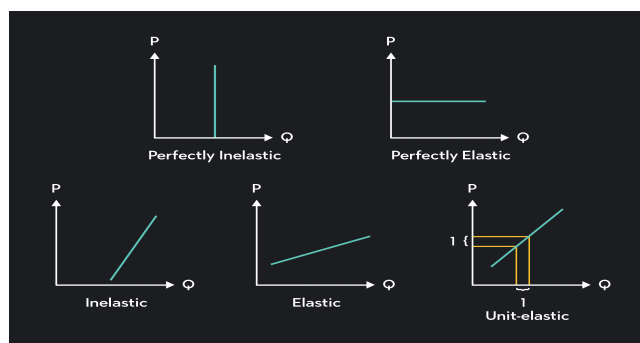


Elasticity of Supply:

Percentage change in quantity supplied due to percentage change in price.

Math: $E_s = (\Delta Q_s / \Delta P) \times (P / Q_s)$

- **Elastic Supply ($E_s > 1$):** Flatter upward curve.
- **Inelastic Supply ($E_s < 1$):** Steeper upward curve.
- **Unit Elasticity ($E_s = 1$):** 45-degree angle starting from origin.
- **Zero Elasticity / Perfectly Inelastic ($E_s = 0$):** Perfectly vertical line.
- **Infinite Elasticity / Perfectly Elastic ($E_s = \infty$):** Perfectly horizontal line.



Chapter 3: Theory of Firm

Lecture-9: 18.08.2020



Production (উৎপাদন): Production is a process that involves transformation of inputs such as land, labor, capital goods and other raw materials into output such as goods and services.

Production Function (উৎপাদন অপেক্ষক)

A function which shows the technical relationship between inputs and outputs.

$$Q = f(L, L_b, K, R, T, t)$$

- L = Land
- L_b = Labor
- K = Capital
- R = Raw materials
- T = Technology
- t = Time

Factors of Production (উৎপাদনের উপকরণ):

1. **Land:** Refers to the surface of land and also all gifts of nature such as rivers, oceans, climate, fisheries, forests.
2. **Labor:** Refers to all mental and physical work undertaken for some monetary unit.
3. **Capital:** Means all man-made resources.
4. **Entrepreneur Skill (Organization):** Refers to the services of an entrepreneur who controls and manages the policy of firms and undertaken all possible risk.

Cost of Production

Lecture-10: 20.08.2020

- **Total Cost (TC):** The sum of all cost for different factors of production. $TC = TFC + TVC$
- **Average Cost (AC):** Total cost divided by output. $AC = TC / Q$
- **Marginal Cost (MC):** Additional cost for producing additional unit of output. $MC = \Delta TC / \Delta Q$

Total cost is divided into two categories:

1. **Total Fixed Cost (TFC):** Expenses which do not change with the change in output/production (e.g., Rent, Interest, wages of permanent workers).
2. **Total Variable Cost (TVC):** Expenses which change with the change in output/production (e.g., raw materials, temporary labor, transport cost).

Cost Calculations and AC/MC Relationship

Lecture-11: 02.09.2020

Relationship between AC and MC:

- **1st Stage:** Both AC and MC are decreasing but AC decreases greater than MC. So, $AC > MC$.
- **2nd Stage:** When AC reaches its minimum point at desired level of production, then $AC = MC$ at point E.
- **3rd Stage:** Both AC and MC are increasing but MC increases greater than AC. So, $MC > AC$.

Comprehensive Cost Calculation Table

Q	FC (Tk)	VC (Tk)	TC = FC+VC	AC = TC/Q	MC = $\Delta TC / \Delta Q$	AFC = FC/Q	AVC = VC/Q
0	100	0	100	-	-	-	-
1	100	60	160	160	60	100	60
2	100	80	180	90	20	50	40
3	100	95	195	65	15	33.33	31.67
4	100	100	200	50	5	25	25
5	100	105	205	41	5	20	21
6	100	120	220	36.67	15	16.67	20
7	100	140	240	34.28	20	14.28	20
8	100	170	270	33.75	30	12.5	21.25
9	100	200	300	33.33	30	11.11	22.22
10	100	240	340	34	40	10	24

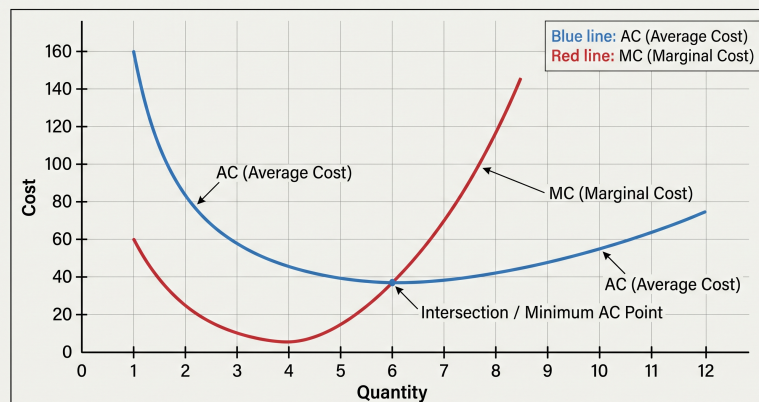
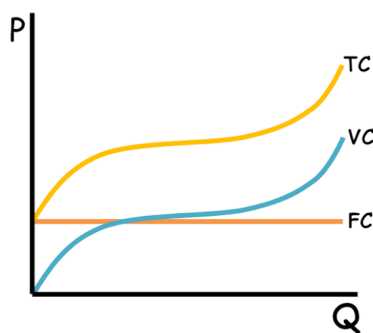


Figure: Relationship Between Average Cost (AC) and Marginal Cost (MC)

Chapter 3

Production and Cost and Theory of the Firm

Production: In economics, production is the process of transforming inputs (like labor, capital, and land) to outputs such as goods and services. Production means creating the utility of a commodity.

Production Function: Production function means the technical relationship between inputs and outputs in the process of production.

The standard equation for a production function is expressed as $Q = f(L, K)$.

Where,

Q = Total output

f = Functional relationship

L = Labor

K = Capital

In terms of time or period, production function can be divided into two categories: (a) short-run production function, (b) long-run production function.

(a) Short-run production function: In the short-run production function at least one factor of production remains fixed.

$Q = f(L, K^0)$, where K is fixed.

(b) Long-run production: In the long-run production all factors of production are variable.

$Q = f(L, K)$. Where both labor (L) and capital (K) change.

Short-run vs long-run production function:

SR	LR
1. In the short-run production function at least one factor of production remains fixed.	1. In the long-run production all factors of production are variable.
2. Limited flexibility	2. High flexibility
3. $Q = f(L, K^0)$	3. $Q = f(L, K)$
4. Factor ratio changes	4. Factor ratio doesn't change
5. Follow law of diminishing returns	5. Follow law of returns to scale.

Factors of production:

- Land: Natural resources such as surface, mineral, air, rivers, sea etc.
- Labor: Mental or physical effort done by a man with the view of producing output.

- Capital: Man-made goods used in the production process
- Organization: Entrepreneur or coordinator of all factors of production.

Cost of production: Cost of production means the total amount a firm spends to make goods and services.

Cost function: The cost of production is a mathematical formula that shows the total amount needed to produce a specific output. It is written as

$$C = f(Q)$$

Where, total cost (C) depends directly on the quantity (Q) of goods produced.

Short-run cost function: In short-run, cost function is indicated by fixed cost and variable cost.

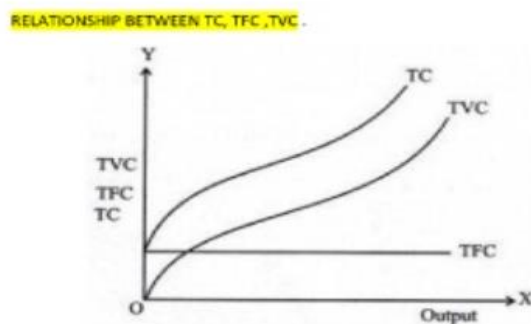
$$C = f(Q) + b; \text{ where } C = \text{Cost}, f(Q) = \text{variable cost}, b = \text{fixed cost}$$

Long-run cost function: In the long run, all costs of production are variable.

$$C = f(Q); \text{ where } C = \text{Cost}, f(Q) = \text{variable cost}$$

Total Cost (TC): The total expenditure incurred by an organization on the factors of production which are required for the production of a commodity is known as Total Cost. In simple terms, total cost is the sum of total fixed cost and total variable cost at different output levels.

$$TC = TFC + TVC$$



Total Fixed cost: All the costs which are not related to the quantity of production, i.e., the costs which do not change with the quantity of production, increase or decrease, but remain constant, are called fixed costs. Generally, house rent, rent, interest on long-term loans, salaries of senior officials, etc. are included in fixed costs.

Total Variable Cost: The cost that changes as a result of a change in the production volume of an organization is called variable cost. If the production volume increases the amount of variable cost increases and if the production volume decreases, this type of cost decreases. For example, wages of workers, cost of purchasing raw materials, electricity bills, etc. are included in variable costs.

Average Cost (AC): The average cost is obtained by dividing the total cost of production by the quantity.

$$\text{Average Cost (AC)} = \frac{\text{Total Cost (TC)}}{\text{Quantity (Q)}}$$

Marginal Cost (MC): The additional cost added to the total cost of producing one additional unit of a particular good is called marginal cost.

$$\text{Marginal Cost (MC)} = \frac{\text{Change in Total Cost } (\Delta TC)}{\text{Change in Quantity } (\Delta Q)}$$

Average Fixed Cost (AFC): Total fixed cost (TFC) is divided by the quantity of production to get average fixed cost (AFC). That is, AFC.

$$\text{Average Fixed Cost (AFC)} = \frac{\text{Total Fixed Cost (TFC)}}{\text{Quantity (Q)}}$$

Average Variable Cost (AVC): Total variable cost (TVC) divided by the quantity of production is the average variable cost (AVC)

$$\text{Average Variable Cost (AVC)} = \frac{\text{Total Variable Cost (TVC)}}{\text{Quantity (Q)}}$$

Relationship between AC and MC:

Average cost is the ratio of total production costs to the total quantity produced.

$$\text{Average Cost (AC)} = \frac{\text{Total Cost (TC)}}{\text{Quantity (Q)}}$$

On the other hand, the amount by which total cost increases to produce one additional unit is marginal cost (MC).

$$\text{Marginal Cost (MC)} = \frac{\text{Change in Total Cost } (\Delta TC)}{\text{Change in Quantity } (\Delta Q)}$$

There is a close relationship between average cost (AC) and marginal cost (MC). This is explained in three stages as follows:

- In the first stage, the initial stage of production, AC and MC usually decrease with the increase in production due to internal and external cost adjustments. However, the rate

at which AC decreases is greater than MC. Therefore, at this stage, average cost (AC) is greater than marginal cost (MC), i.e., $AC > MC$.

- In the second stage, when average cost (AC) reaches its minimum at the desired level of production, average cost and marginal cost are equal to each other. That is, $AC = MC$

- In the third stage, after production reaches the minimum level, if production is increased further, average and marginal cost continue to increase. However, the growth rate of marginal cost is higher than average cost. Therefore, after the transition, average cost is less than marginal cost, i.e., $AC < MC$.

The relationship between AC and MC is explained in the diagram.

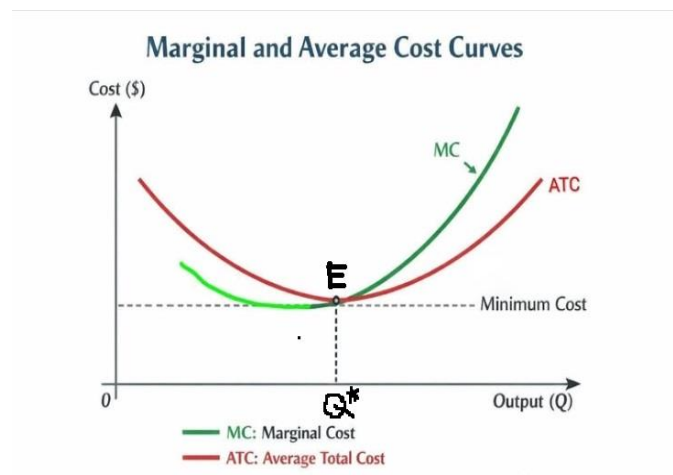


Figure: AC and MC

In the figure, AC and MC are the average and marginal cost curves respectively. Until the quantity OQ is produced, the AC line gradually slopes downwards to the right. As long as the AC line slopes downwards, the MC line remains below the AC line, i.e., at this stage, the quantity of production $AC > MC$ is OQ. At this OQ production level, at point E . i.e., at the lowest point of the AC line, the MC line crosses the AC line from below and rises upwards. Therefore, at the lowest point of the AC line, $MC = AC$ has become. After the production of OQ, both AC and MC slope upwards to the right and the MC line is above the AC line because the rate of increase in marginal cost is higher than the average cost. That is, after the production of OQ, $MC > AC$ will be. The AC line slopes downwards to point E and then upwards. E is the lowest point of the AC line. Therefore, since it slopes downwards first and then upwards, the shape of the AC and MC lines is like the English letter "U".

Using the following table compute the Total Cost (TC), Marginal Cost (MC), Average Cost (AC), Average Variable Cost (AVC), Average Fixed Cost (AFC).

Q	FC	VC	TC	AC	MC	AFC	AVC
0	100	0					
1	100	60					
2	100	80					
3	100	95					
4	100	100					
5	100	105					
6	100	120					
7	100	140					
8	100	170					
9	100	200					
10	100	210					